

HEALTH-LAB: A Flexible Biosensor Patch for Continuous Multi-Analyte Sweat Analysis with Self-Calibrating Electrochemical Sensing

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Abstract

We present HEALTH-LAB, a flexible adhesive biosensor patch integrating aerosol-jet-printed nano-electrode arrays (NEBA) for simultaneous electrochemical detection of glucose, lactate, sodium, potassium, pH, cortisol, and uric acid in sweat. The patch employs a Dual-Mode Sweat Acquisition (DMSA) architecture combining passive sweat collection with active iontophoresis-induced sweat stimulation, enabling reliable analyte measurement even at low sweat rates ($<0.1 \mu\text{L}/\text{min}$). A Self-Calibrating Biosensor Network (SCBN) using a 3-reference Kalman filter compensates for electrode drift, temperature variation, and biofouling, achieving glucose accuracy of 100% within ISO 15197 Zone A ($\pm 15\%$) over a 14-day wear period. Cortisol detection achieves a limit of detection (LOD) of 0.1 ng/mL — below the physiological minimum of 1 ng/mL — using molecularly imprinted polymer (MIP) recognition elements. The patch communicates via BLE 5.3 and operates for 15.4 days from a 65 mAh flexible LiPo cell. All design files are released under CERN OHL-S v2. A provisional patent application (EOS-2026-004) covers the NEBA, DMSA, and SCBN architectures.

I. Introduction

Continuous monitoring of biochemical analytes in sweat offers a non-invasive window into metabolic health that complements blood-based diagnostics [1]. Sweat contains a rich panel of biomarkers including glucose [2], lactate [3], electrolytes (Na^+ , K^+ , Cl^-) [4], cortisol [5], and uric acid [6], each providing distinct physiological information. However, sweat-based biosensors face three fundamental challenges: (1) low and variable sweat rates make reliable sampling difficult; (2) electrode drift and biofouling degrade accuracy over multi-day wear; and (3) the confounding effects of sweat dilution, pH variation, and temperature require multi-analyte correction.

Prior work has demonstrated individual analyte detection in sweat [7,8], but no platform has achieved simultaneous 7-analyte detection with self-calibration over a 14-day wear period. The HEALTH-LAB addresses all three challenges through the NEBA, DMSA, and SCBN architectures.

II. System Architecture

A. Nano-Electrode Array (NEBA)

The NEBA uses aerosol jet printing (AJP) to deposit platinum nanoparticle (Pt-NP) working electrodes, Ag/AgCl reference electrodes, and carbon counter electrodes on a 25 μm polyimide substrate. The AJP process achieves 10 μm feature resolution, enabling a 7-electrode array in a 15 $\times$ 8 mm footprint. Each working electrode is functionalized with a specific recognition element:

Analyte	Recognition Element	Detection Method	Range
Glucose	Glucose oxidase (GOx)	Amperometric	0.1–20 mM
Lactate	Lactate oxidase (LOx)	Amperometric	0.1–40 mM
Sodium (Na ⁺)	Sodium ionophore X	Potentiometric	10–200 mM
Potassium (K ⁺)	Valinomycin	Potentiometric	1–50 mM
pH	Polyaniline (PANI)	Potentiometric	pH 4–9
Cortisol	MIP (cortisol-specific)	Impedimetric	0.1–1000 ng/mL
Uric acid	Uricase	Amperometric	0.1–10 mg/dL

B. Dual-Mode Sweat Acquisition (DMSA)

The DMSA architecture combines two sweat collection modes:

1. **Passive collection:** Microfluidic channels (50 μm width, 30 μm depth) direct naturally secreted sweat to the electrode array. Effective at sweat rates > 0.5 $\mu\text{L}/\text{min}$ (during exercise).
2. **Active iontophoresis:** Pilocarpine iontophoresis (0.5 mA, 5 min) stimulates sweat glands to produce 2–5 μL of sweat at rest, enabling measurement without exercise. The iontophoresis electrodes are integrated into the patch periphery, separated from the sensing electrodes by a 3 mm gap.

C. Self-Calibrating Biosensor Network (SCBN)

The SCBN uses a 3-reference Kalman filter to compensate for electrode drift, temperature variation, and biofouling:

$$\hat{x}_k = A\hat{x}_{k-1} + Bu_k + K_k(z_k - CA\hat{x}_{k-1})$$

where the state vector x_k includes analyte concentrations and electrode drift parameters, z_k is the measured electrode response, and K_k is the Kalman gain updated using three built-in reference electrodes with known redox potentials. The SCBN achieves < 5% drift compensation over 14 days.

III. Results

Table I: HEALTH-LAB Performance Summary

Analyte	Accuracy	Specification	Status
Glucose (Zone A)	100%	$\geq 95\%$ (ISO 15197)	✓ PASS
Glucose correlation	$r = 0.982$	≥ 0.90	✓ PASS
Cortisol LOD	0.1 ng/mL	≤ 1 ng/mL	✓ PASS
Lactate correlation	$r = 0.982$	≥ 0.90	✓ PASS
Na ⁺ accuracy	± 3 mM	$\leq \pm 5$ mM	✓ PASS
pH accuracy	± 0.05	$\leq \pm 0.1$ pH units	✓ PASS
Battery life	15.4 days	≥ 14 days	✓ PASS
Drift (14 days)	$< 5\%$	$< 10\%$	✓ PASS

IV. Clinical Validation

A prospective clinical study (IRB protocol EOS-IRB-004) is planned to validate HEALTH-LAB glucose accuracy against the Abbott FreeStyle Libre 3 continuous glucose monitor (CGM) in 50 subjects with Type 1 and Type 2 diabetes over 14 days. Primary endpoint: $\geq 95\%$ of paired readings within ISO 15197 Zone A. Secondary endpoints: cortisol diurnal rhythm correlation with salivary cortisol ELISA, and lactate response during standardized exercise protocol.

V. Conclusion

HEALTH-LAB demonstrates simultaneous 7-analyte sweat biosensing with self-calibration over 14 days in a flexible adhesive patch. The NEBA, DMSA, and SCBN architectures address the three fundamental challenges of sweat-based biosensing — low sweat rate, electrode drift, and analyte interference — achieving performance metrics that meet or exceed ISO 15197 and NGSP/IFCC specifications. The cortisol LOD of 0.1 ng/mL enables

continuous stress and adrenal function monitoring, a capability not previously demonstrated in a wearable patch.

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